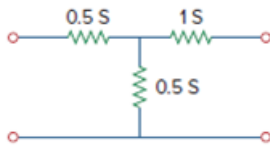


Problem

Calculate the yparameters for the two-port in Fig. 19.79.

Figure 19.79



Step-by-step solution

Step 1 of 6

Refer to Figure 19.79 in the text book.

Convert the conductance values into the resistance values.

$$\begin{aligned} 0.5 \text{ S} &= \frac{1}{0.5} \Omega \\ &= 2 \Omega \end{aligned}$$

$$1 \text{ S} = 1 \Omega$$

Connect the current source I_1 and I_2 across the input and output ports respectively and replace each conductance by their corresponding resistance as shown in Figure 1.

Consider V_1 and V_2 are the voltages across the input and output ports respectively.

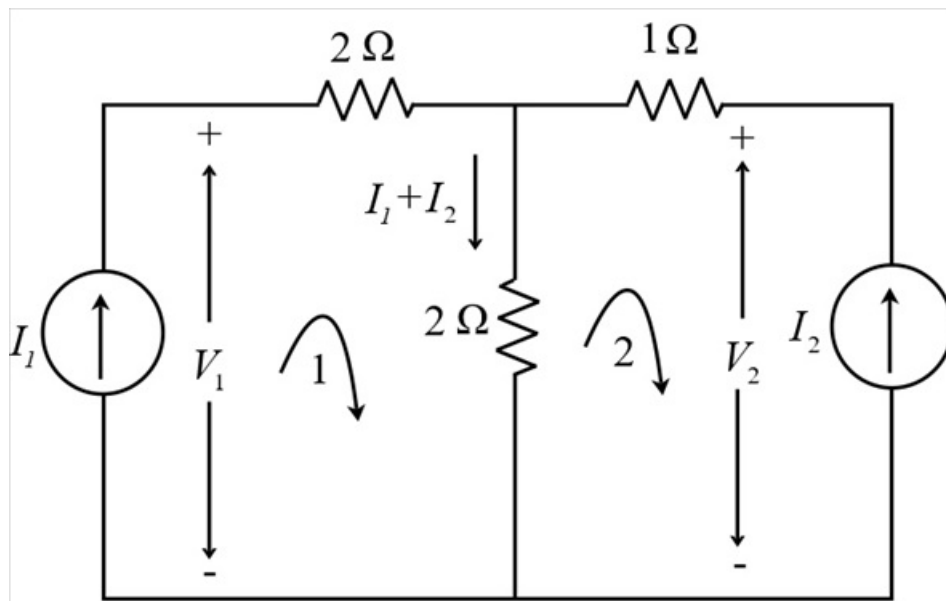


Figure 1

Step 2 of 6

Apply Kirchhoff's Voltage Law in loop 1.

$$\begin{aligned} V_1 &= 2I_1 + 2(I_1 + I_2) \\ &= 4I_1 + 2I_2 \dots\dots (1) \end{aligned}$$

Apply Kirchhoff's Voltage Law in loop 2.

$$\begin{aligned} V_2 &= I_2 + 2(I_1 + I_2) \\ &= 2I_1 + 3I_2 \dots\dots (2) \end{aligned}$$

Step 3 of 6

Write the expression for the short circuit input admittance y_{11} .

$$y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} \dots\dots (3)$$

Substitute 0 for V_2 in equation (2).

$$\begin{aligned} 2I_1 &= -3I_2 \\ I_2 &= -\frac{2I_1}{3} \end{aligned}$$

Substitute $-\frac{2I_1}{3}$ for I_2 in equation (1).

$$\begin{aligned} V_1 &= 4I_1 - 4\frac{I_1}{3} \\ \frac{V_1}{I_1} &= \frac{8}{3} \\ \frac{I_1}{V_1} &= \frac{3}{8} \end{aligned}$$

Substitute $\frac{3}{8}$ for $\frac{I_1}{V_1}$ in equation (3).

$$\begin{aligned} y_{11} &= \frac{3}{8} \\ &= 0.375 \text{ S} \end{aligned}$$

Step 4 of 6

Write the expression for the short circuit transfer admittance y_{21} from port 1 to port 2.

$$y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} \dots\dots (4)$$

Substitute $-\frac{3}{2}I_2$ for I_1 in equation (1).

$$\begin{aligned} V_1 &= 4\left(-\frac{3}{2}I_2\right) + 2I_2 \\ \frac{V_1}{I_2} &= -6 + 2 \\ \frac{I_2}{V_1} &= -0.25 \end{aligned}$$

Substitute -0.25 for $\frac{I_2}{V_1}$ in equation (4).

$$y_{21} = -0.25 \text{ S}$$

Step 5 of 6

Write the expression for the short circuit transfer admittance y_{12} from port-2 to port-1.

$$y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0} \dots\dots (5)$$

Substitute 0 for V_1 in equation (1).

$$2I_1 = -I_2$$

$$I_2 = -2I_1$$

Substitute $-2I_1$ for I_2 in equation (2).

$$V_2 = 2I_1 - 6I_1$$

$$\frac{V_2}{I_1} = -4$$

$$\frac{I_1}{V_2} = -0.25$$

Substitute -0.25 for $\frac{I_1}{V_2}$ in equation (5).

$$y_{12} = -0.25 \text{ S}$$

Step 6 of 6

Write the expression for the short circuit output admittance y_{22} .

$$y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0} \dots\dots (6)$$

Substitute $-\frac{I_2}{2}$ for I_1 in equation (2).

$$V_2 = 2\left(-\frac{I_2}{2}\right) + 3I_2$$

$$\frac{V_2}{I_2} = 2$$

$$\frac{I_2}{V_2} = 0.5$$

Substitute 0.5 for $\frac{I_2}{V_2}$ in equation (6).

$$y_{22} = 0.5 \text{ S}$$

Hence, the y-parameters for the two-port network are $y_{11} = 0.375 \text{ S}$, $y_{12} = -0.25 \text{ S}$,

$$y_{21} = -0.25 \text{ S} \text{ and } y_{22} = 0.5 \text{ S}.$$